

07.3 — Admin User Guide

For: Venkat (system administrator) and any operator with superuser access to Haritha Hospitals ERPNext. **Goal:** Maintain, troubleshoot, upgrade, and customize the system. Reference-style — read what you need. **Last updated:** 2026-08-29
Pre-req reading: 00.2 Architecture Overview, 04.1 Deployment Runbook, 04.2 Daily Ops Runbook, 04.3 Disaster Recovery, 04.4 Incident Response

System overview

The Haritha Hospitals ERPNext system runs on a single VPS (main + offsite) with three environments:

Env	URL	Purpose
pberpdev	https://pberpdev.duckdns.org/	Development — schema experiments, fixture testing
pberpqa	https://pberpqa.duckdns.org/	Pre-prod gate — currently skipped for Haritha work
pberpprod	https://pberpprod.duckdns.org/	Production — live users

Each env is a Docker Compose stack:

- Backend (gunicorn, Frappe workers)
- DB (MariaDB)
- Scheduler (cron jobs, scheduled tasks)
- Queue (background jobs)

Customizations live in the `haritha_hospital` app (workspace: `projects/haritha-hospitals/`, repo: `git@github.com:venkat-narasimha/haritha_hospital.git`). As of 2026-08-29, the app captures 274 customizations: 78 Custom Fields + 189 Property Setters + 3 Print Formats + 2 Notifications + 2 Letter Heads.

Architecture recap: see 00.2 Architecture Overview.

Environment management

3-env setup

Env	DB Name (in <code>site_config.json</code>)	DB Password Location	SSH User
pberpdev	<code>_0c0679ad719b9491</code>	<code>MYSQL_ROOT_PASSWORD</code> env on <code>erp-dev-db-1</code> container	vijay
pberpqa	<code>_9ffca8aa07bbfdcc</code>	<code>MYSQL_ROOT_PASSWORD</code> env on <code>erp-qa-db-1</code> container	vijay

Env	DB Name (in site_config.json)	DB Password Location	SSH User
pberpprod	_b80f05e76a0dcaad	MYSQL_ROOT_PASSWORD env on erp-prod-db-1 container	vijay

Memory note (LEARNINGS #154, #157): DB passwords in MEMORY.md can drift. **Always re-verify** via:

```
ssh -i /root/.openclaw/ssh_key vijay@144.217.163.228 \
  "docker exec erp-prod-db-1 printenv MYSQL_ROOT_PASSWORD"
```

The container env is the source of truth. Update MEMORY.md quarterly.

When to use which

Scenario	Use
Schema experiment, custom field add, fixture test	pberpdev
Pre-prod validation before production deploy	pberpqa (currently light use)
Smoke testing after deploy	pberpprod
User-reported issue triage (read-only)	pberpprod
Anything destructive (delete, schema change)	pberpdev first, then promote

Rule: Never run bench migrate, custom field create, or install-app directly on prod. Always go dev → QA → prod.

SSH access patterns

```
# Main VPS (vijay, hosts all 3 envs)
ssh -i /root/.openclaw/ssh_key vijay@144.217.163.228

# Offsite backup VPS (venkat)
ssh -i /root/.openclaw/venkat_vps_key venkat@135.125.196.35

# Inside container (after SSH to host)
docker exec -it erp-prod-backend-1 bash
docker exec -it erp-prod-db-1 bash
```

SSH key per host. Don't reuse keys. See "Security" below for rotation.

DB password locations (per env)

```
# Read DB password from container env (source of truth)
for env in dev qa prod; do
  echo "=== $env ==="
  ssh vijay@144.217.163.228 \
    "docker exec erp-${env}-db-1 printenv MYSQL_ROOT_PASSWORD"
done
```

If you need to set a new DB password (rare — see DB password rotation), update container env in `docker-compose.yml`, restart db container, then update `sites/<site>/site_config.json`.

App management

haritha_hospital app — what it does

The custom app captures all Haritha-specific customizations as code:

- 78 Custom Fields (e.g. `Shift Type.is_oncall`, `Employee.blood_group`)
- 189 Property Setters (e.g. `Shift Type.color = #hex` for roster SPA colors)
- 3 Print Formats (Salary Slip, Leave Application, Offer Letter)
- 2 Notifications (leave approval, shift swap)
- 2 Letter Heads (Haritha Hospitals branded PDFs)

The app is **env-portable**: install on a fresh env → all 274 customizations apply automatically. No more “prod has a field dev doesn’t” drift.

Update flow (dev → prod)

1. **Dev** (local workspace):
 - Edit `haritha_hospital/fixtures/*.json` or `haritha_hospital/hooks.py`
 - Test on `pberpdev`: `bench --site pberpDEV.duckdns.org console + UI verify`
 - Commit: `git add . && git commit -m "<msg>"` (identity: `venkat-narasimha / srivenkatnarasimha@gmail.com`)
2. **Promote to QA** (`pberpqa`):
 - `ssh vijay@144.217.163.228 "cd /home/vijay/frappeclaw/frappeclaw-data/workspace/projects/haritha_hospital && git pull"`
 - `bench --site pberpqa.duckdns.org migrate`
 - Smoke test
3. **Promote to Prod**:
 - Same `git pull + bench --site pberpprod.duckdns.org migrate`
 - **Always** `docker restart erp-prod-backend-1` after install-app or major code change (LEARNINGS #153)

Fixtures re-tag workflow (when adding new customizations)

If you create a new Custom Field in `pberpprod` UI:

1. **Don’t leave it there** — it becomes prod-only drift.
2. Re-tag it to the `haritha_hospital` module:
 - Custom Field → edit → set `Module = haritha_hospital` → save
3. Re-export fixtures:
`bench --site pberpprod.duckdns.org export-fixtures --app haritha_hospital`
4. Verify `haritha_hospital/fixtures/custom_field.json` includes the new field.
5. Commit + push.

Hooks requirement (LEARNINGS #151): `fixtures = [...]` must be declared in `haritha_hospital/hooks.py`. Without it, `export-fixtures` exits 0 with no files written (silent no-op). Current declaration:

```

fixtures = [
  {"doctype": "Custom Field", "filters": {"module": "haritha_hospital"}},
  {"doctype": "Property Setter", "filters": {"module": "haritha_hospital"}},
  {"doctype": "Print Format", "filters": {"module": "haritha_hospital"}},
  {"doctype": "Notification", "filters": {"module": "haritha_hospital"}},
  {"doctype": "Letter Head", "filters": {"module": "haritha_hospital"}},
]

```

Customization

Adding a Custom Field (UI + code)

UI method (quick, for one-off):

1. Settings → Customize → Custom Field → New.
2. Pick DocType (e.g. Employee).
3. Add field: label, fieldname, fieldtype, options.
4. **Module = haritha_hospital** (critical for fixture capture).
5. Save.
6. Re-export fixtures (see above).

Code method (for version control):

1. Edit haritha_hospital/fixtures/custom_field.json.
2. Add entry with all required fields:

```

{
  "doctype": "Custom Field",
  "dt": "Employee",
  "fieldname": "blood_group",
  "label": "Blood Group",
  "fieldtype": "Select",
  "options": "A+\nA-\nB+\nB-\nAB+\nAB-\nO+\nO-",
  "module": "haritha_hospital",
  "insert_after": "emergency_contact"
}

```

3. Commit + push.
4. `bench --site pberpprod.duckdns.org migrate` (loads fixtures).

Adding a Property Setter

Property Setters modify stock fields (change field type, hide, set color, etc.). Example — set Shift Type color:

1. Settings → Customize → Property Setter → New.
2. DocType = Shift Type, Property = color, Value = #FF8C00 (orange for Afternoon).
3. Module = haritha_hospital.
4. Save + re-export fixtures.

Common uses:

- **color** on Shift Type (drives roster SPA)
- **hidden** to hide a stock field
- **options** to narrow a Select

- **read_only** to lock a field from non-admin edits
- **default** to prefill values

Adding a Print Format

1. Settings → Printing → Print Format → New.
2. Pick DocType (e.g. Salary Slip).
3. Write Jinja-templated HTML (or use the UI builder).
4. Module = haritha_hospital.
5. Add to fixtures (print_format.json).

Test: Preview with sample data → PDF → verify letter head + layout → commit.

Exporting as fixtures

```
bench --site <site> export-fixtures --app haritha_hospital
```

Files written to haritha_hospital/fixtures/:

- custom_field.json
- property_setter.json
- print_format.json
- notification.json
- letter_head.json
- (any other DocTypes in hooks.py:fixtures)

Verifying fixtures

Count customizations per file

```
jq 'length' haritha_hospital/fixtures/custom_field.json # expect 78
jq 'length' haritha_hospital/fixtures/property_setter.json # expect 189
jq 'length' haritha_hospital/fixtures/print_format.json # expect 3
jq 'length' haritha_hospital/fixtures/notification.json # expect 2
jq 'length' haritha_hospital/fixtures/letter_head.json # expect 2
```

Sanity: all CF/PS rows tagged correctly

```
jq '[.[] | select(.module != "haritha_hospital")]' haritha_hospital/fixtures/custom_field.json
# Should be empty []
```

User & permission management

Creating users

1. Settings → User → New.
2. Email (must be unique).
3. First Name, Last Name.
4. **Roles** child table — add appropriate roles (Employee, HR Manager, HR User, etc.).
5. **Module Profile** — optional, bundles roles.
6. Save.
7. System sends invite email with temp password.

Test users (e.g. for QA): use +test1@harithahospitals.in aliases; do not pollute real user list.

Role assignment

Built-in roles (used at Haritha):

- **Administrator** — full access
- **System Manager** — settings, users, permissions (no transactions)
- **HR Manager** — full HR module
- **HR User** — read-only HR + own records
- **Employee** — self-service only
- **Employee Self Service** — subset of Employee (limited views)
- **Accounts Manager, Accounts User** — finance (not used currently)
- **Leave Approver** — can approve leaves only

Custom roles

For fine-grained access:

1. Settings → Role → New.
2. Name (e.g. Nurse Roster Editor).
3. Description.
4. Permissions: child table (DocType + level).
5. Save.

Custom roles rarely needed. Reuse built-in roles where possible.

Permission rules (DocType-level, field-level)

DocType-level (Settings → Role Permissions):

- For each role × DocType combo: read / write / create / delete / submit / cancel / amend flags.

Field-level (rare):

- Property Setter with property = permlevel or custom perm rules.

Row-level (User Permissions):

- Settings → User Permission → assign what records a user can see.
- Example: HR manager can only see Employees in Department = Cardiology.

Verify: After any role/permission change, log in as that user (or impersonate via bench --site X console frappe.session.user = 'X') and check access.

Master data

Departments, Designations, Shift Types, Locations

Naming conventions (critical for consistency — see masters/shift_type.csv):

- **Department:** Title Case, full name (Cardiology, Pediatrics, Emergency Medicine)
- **Designation:** Title Case (Staff Nurse, Senior Consultant, Resident Doctor)
- **Shift Type:** [Period][HHMM][R/S][HHMM]
 - Period: G (General), M (Morning), A (Afternoon), N (Night)
 - HHMM = 24-hour start time
 - R = Regular (single block), S = Splits past midnight
 - Last HHMM = end time

- Example: M0800R0830 = Morning 08:00, Regular, ends 08:30 next day = 24.5h ... actually 08:00 → 16:30 = 8.5h
- **Location:** Building / Floor / Wing (Main Block, ICU 1st Floor, OP Wing)

Holiday Lists

1. Human Resources → Holiday List → New.
2. Add holidays row-by-row (date, name).
3. Set weekly offs (typically Sundays).
4. Assign to Department or Employee via Employee record → Holiday List field.

Single source of truth: “Haritha Hospitals Holiday List” — all departments share. No per-department lists unless absolutely needed.

Backup & recovery

See 04.3 Disaster Recovery Runbook for full procedure.

Quick reference:

- **Local backups:** /home/vijay/backups/{prod,qa,dev}/ (7d retention)
 - **Offsite backups:** venkat@135.125.196.35:/home/venkat/pberp{prod,qa,dev}_backups/ (forever)
 - **Schedule:** 4x/day (00:00, 06:00, 12:00, 18:00 IST)
 - **Verify:** sha256sum local vs offsite daily (per Daily Ops)
-

Performance monitoring

Container health

```
# All 4 envs, all containers
ssh vijay@144.217.163.228 \
  "docker ps --format 'table {{.Names}}\t{{.Status}}\t{{.Ports}}' \
  | grep -E 'erp-(prod|qa|dev)-(backend|db|scheduler|queue)-1'"

# Look for: RestartCount > 0, Up X minutes (not hours), Exited state
docker inspect <container> --format '{{.State.RestartCount}} {{.State.StartedAt}}'
```

See 04.2 Daily Ops Runbook for daily cadence.

DB queries (slow query log)

```
# Enable slow query log
docker exec erp-prod-db-1 bash -c \
  "mysql -uroot -p\"$MYSQL_ROOT_PASSWORD\" -e \
  \"SET GLOBAL slow_query_log = 'ON'; \
  SET GLOBAL long_query_time = 2;\""

# Tail slow log
docker exec erp-prod-db-1 tail -f /var/log/mysql/slow.log

# Disable after session
docker exec erp-prod-db-1 mysql -uroot -p\"$MYSQL_ROOT_PASSWORD\" \
  -e "SET GLOBAL slow_query_log = 'OFF';"
```

Default threshold: 2 seconds. Adjust per need.

Worker count tuning

Edit `frappe_docker/compose.yml` or `common_site_config.json`:

```
{
  "workers": {
    "background_workers": 4,
    "short_workers": 4,
    "long_workers": 4
  }
}
```

After change: `docker compose up -d` (full restart of worker containers).

Rule of thumb: 1 worker per ~25 concurrent users. Haritha has ~250 employees, ~80 concurrent users → 4 workers each is enough.

Troubleshooting

Common errors + fixes

Error	Likely cause	Fix
500 Server Error after install-app	gunicorn --preload sys.path not refreshed	<code>docker restart erp-prod-backend-1</code> (LEARNINGS #153)
ImportError: No module named haritha_hospital Permission denied for routine action Login fails with 401	app not in <code>apps.txt</code> or no backend restart User missing role	Check <code>apps.txt</code> , restart backend Settings → User → add role
Connection refused to DB bench backup exit 1	DB password mismatch	Re-verify via <code>printenv</code> (LEARNINGS #154, #157)
Roster SPA blank	DB container down hrms ghost in <code>installed_apps</code> (Lesson #57)	<code>docker ps</code> ; restart if down Edit tabInstalled Application row
Print Format missing	Custom Field not loaded Letter Head deleted	<code>bench migrate</code> ; verify <code>haritha_hospital</code> installed Restore from fixtures; re-tag module

See 04.4 Incident Response Plan for full triage flow.

Reference **LEARNINGS.md** for known issues

`/root/.openclaw/workspace/.learnings/LEARNINGS.md` has 100+ lessons. Key ones for Haritha:

- #42 — Shift Type `last_sync_of_checkin` required for auto-attendance
- #48 — 8-phase test playbook (mandatory after any install)
- #52-58 — HRMS vendor test bootstrap blockers

- #79 — backup SHA256 integrity check
- #151 — export-fixtures requires hooks declaration
- #153 — backend restart after install-app
- #154, #157 — DB password drift in MEMORY

Always grep LEARNINGS.md before debugging a known pattern. Saves 30+ min per session.

Upgrades

Frappe / ERPNext / HRMS upgrade procedure

1. **Backup first** (always):
`bench --site <site> backup --with-files`
2. **Pull new image:**
`cd /home/vijay/frappeclaw`
`docker compose pull`
3. **Run migrations:**
`bench --site <site> migrate`
4. **Restart all:**
`docker compose restart`
5. **8-phase test** (per 08.3 Regression Test Script)
6. **Smoke test in browser:** login, load key DocTypes.

haritha_hospital compatibility check

Before any upgrade:

```
cd /home/vijay/frappeclaw/frappeclaw-data/workspace/projects/haritha_hospital
git log --oneline -10 # recent commits
git status # clean?
```

Check if any fixtures reference deprecated fields. Grep fixtures/*.json for the fieldname before migrating.

Pre-upgrade checklist

- ☐ Backup taken + SHA256 verified
 - ☐ haritha_hospital on clean commit, no uncommitted changes
 - ☐ No active long-running jobs (pause scheduler)
 - ☐ No users logged in during upgrade window
 - ☐ Maintenance window communicated (24h notice)
 - ☐ Rollback plan documented (last working image tag)
-

Security

DB password rotation

When: quarterly or after any suspected leak.

1. Backup

```
bench --site pberpprod.duckdns.org backup --with-files
```

2. Edit docker-compose.yml → MYSQL_ROOT_PASSWORD=newpassword

```
ssh vijay@144.217.163.228 "vim /home/vijay/frappeclaw/docker-compose.yml"
```

3. Restart db container

```
ssh vijay@144.217.163.228 \  
"cd /home/vijay/frappeclaw && docker compose up -d db"
```

4. Update site_config.json

```
ssh vijay@144.217.163.228 \  
"docker exec erp-prod-backend-1 bash -c \  
\"python -c \\\"import json; \  
  c = json.load(open('/home/frappe/frappe-bench/sites/pberpprod.duckdns.org/site_config.json')); \  
  c['db_password'] = 'newpassword'; \  
  json.dump(c, open('/home/frappe/frappe-bench/sites/pberpprod.duckdns.org/site_config.json', 'w'))\""
```

5. Restart backend

```
ssh vijay@144.217.163.228 "docker restart erp-prod-backend-1"
```

6. Verify login

```
bench --site pberpprod.duckdns.org console
```

→ if console opens, password works

7. Update MEMORY.md

SSH key management

- **One key per host** (no reuse).
- Store in /root/.openclaw/{ssh_key,venkat_vps_key,pberppqa_backup_key,...}.
- chmod 600 on private key.
- Rotate annually.
- Revoke on personnel change.

SSL cert renewal

DuckDNS provides Let's Encrypt certs via certbot in the nginx container:

```
ssh vijay@144.217.163.228 \  
"docker exec erp-prod-nginx-1 certbot renew --dry-run"
```

If dry-run passes:

```
ssh vijay@144.217.163.228 \  
"docker exec erp-prod-nginx-1 certbot renew && docker exec erp-prod-nginx-1 nginx -s reload"
```

Auto-renewal: cron in container checks daily at 03:00. Verify after 60 days.

Audit log review

Periodic checks (monthly recommended):

Recent schema changes

```
bench --site pberpprod.duckdns.org console
```

```
>>> frappe.db.sql("SELECT creation, modified, modified_by FROM `tabCustom Field` WHERE cre
```

```
>>> frappe.db.sql("SELECT creation, modified, modified_by FROM `tabProperty Setter` WHERE
```

User login anomalies

```
>>> frappe.db.sql("SELECT user, MAX(time) FROM `tabActivity Log` WHERE creation > NOW() -
```

Failed logins

```
>>> frappe.db.sql("SELECT * FROM `tabActivity Log` WHERE status='Failed' AND creation > NO
```

Anomaly response: Multiple failed logins from one IP → possible brute force → block IP. New custom fields outside haritha_hospital module → drift → re-tag.

Where to next

- **Day-to-day monitoring?** See 04.2 Daily Ops Runbook
- **Disaster recovery?** See 04.3 Disaster Recovery Runbook
- **Incident response?** See 04.4 Incident Response Plan
- **Testing?** See 08.1 Test Plan and 08.3 Regression Test Script
- **Process changes?** See 05.1 Change Management Policy